

The Century of Building Human Capital Is Over

Research note, September 2026. The 1925-2025 reconstruction is produced by `scripts/human-capital-backcast.py`; the 2025-2100 projection by `npm run human-capital:trajectory` on the model's default parameters, with sensitivity runs noted where used.

Abstract

The world built human capital without interruption for a hundred years and has now stopped. That is the result of pricing people times education as a capital asset: capitalize what it costs to rear and school each workforce entrant, at today's cost, and depreciate it over the years the entrant is expected to spend at work. Between 1925 and 2025 the stock embodied in the working-age population grew between five- and sixfold, 1.7% a year, of which 1.6 points was more people of working age and 0.1 was more education per person. The growth accelerated to 2.5% a year in 1975-2000, slowed to 1.4% in 2000-2025, and to 0.8% in the last decade. In 2025 the ledger capitalizes exactly what it charges: \$15.2T of new entrants against \$15.2T of depreciation and write-offs, 9.6% of world GDP each way.

From here the model's demographics, which follow the UN's low-fertility variant, take net investment negative from the 2040s and to -\$26T a year by

1. The stock of people-times-education still shows a 22% rise to a 2063

peak, but almost none of that is new human capital: through 2050 the gain is \$44T at 2025 cost, of which \$1T is net investment, \$36T is migrants re-priced at their destination's higher replacement cost, and \$7T is the longer working lives of people already at work. With migration switched off the world stock peaks 9% up in the 2050s and ends the century 8% below today. What is being built is education rather than headcount: entrants fall from 119M to 96M a year while the tertiary share of the workforce goes from a quarter to three-fifths.

The regions divide four ways. China and Russia are drawing down now, China by 61% by 2100. India, Latin America, and South-East Asia build to the 2050s and give it back; the Middle East peaks in the 2060s and holds. Sub-Saharan Africa builds throughout. The United States and the rest of the OECD grow only through immigration: the US covers the wear on its own cohorts (its own births, not immigrants) with a small surplus until 2064, and the immigrants it books on top become over a fifth of its workforce by 2075, at which point their own depreciation absorbs the inflow. Spending on new entrants stays near 10% of GDP, where it has been for a century, and slips to 8% by 2100. The ledger is cost, not value; it says nothing about what the people can do, only what it took to make them.

Backcast, 1925-2025

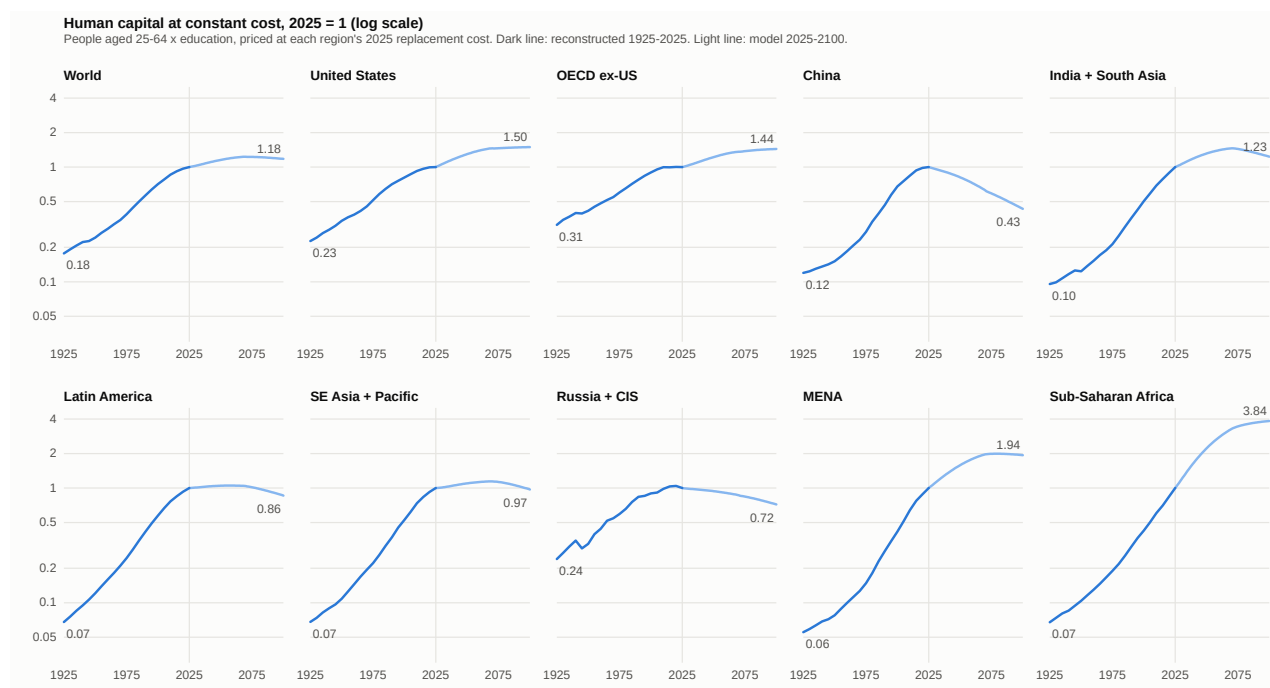
The ledger

The account is the `humanCapital` module documented in `docs/HUMAN_CAPITAL.md`: a Kendrick-style cost-based ledger at current replacement cost. Entrants are sorted into four education bands (primary, secondary, tertiary, advanced), each with its own entry age, and priced as a multiple of the region's GDP per capita: rearing at the USDA out-of-pocket share (23% of GDP per capita a year through the band's entry age) plus OECD

spending per student for each schooling stage. This is the explicit-outlay scope; students' foregone earnings are out of it by default and are a sensitivity below. Depreciation is straight-line over the expected time in the workforce, from a survival curve that nets out death, disability, domestic-role, and retirement exits; workers who leave early are written off at remaining book value. Depreciation plus write-offs is the year's charge, and the people still on the books are the in-service workforce. The ledger is diagnostic: it reads demographics and GDP and feeds nothing back.

Because unit costs track income, the dollar stock rises with GDP per capita even when the workforce it prices is shrinking; on the default path the world net stock goes from \$233T in 2025 to \$2,909T in 2100, a 12-fold rise that is almost entirely revaluation. Every "growing or shrinking" statement in this note therefore deflates each region's stock by that region's own GDP-per-capita index (2025 = 1): a constant-cost quantity, how many 2025-dollars of rearing and schooling are embodied in the people at work.

The simulation starts in 2025 and cannot run backwards, so the century before it is reconstructed from observed data and priced with the same multipliers. The stock is people aged 25-64 times their education (Lee and Lee's long-run attainment series, 1870-2040, and Barro-Lee by age group); entrants are the population aged 20-24 divided by five, with the attainment mix of the cohort five years on. Costs are multiples of GDP per capita (Maddison), so investment and stock relative to GDP depend only on demographics and education, and the constant-cost stock holds each region's 2025 unit cost fixed. Countries are grouped into the model's nine regions so that the 2025 totals match the model's population anchors within a few percent (Sub-Saharan Africa is the exception, at 12% below). Lee-Lee and Barro-Lee count China's completed tertiary attainment at about 3% of ages 25-64 in 2015, far below the census, so China's tertiary shares from 2000 on are overridden with the NBS census communiqués (junior college and above, rescaled to ages 25-64) and the Ministry of Education's gross tertiary enrollment ratio for entrants, ending at the model's own 2025 anchor. Before 1950 the level follows each region's total population from the 1950 age structure, with Lee-Lee supplying the education mix and the change in age shares, because its pre-1950 country series sit on older territories (British India, the USSR). The headline series is gross (no age-based write-down) so that it can reach back to 1925; a net series from the age structure is given from 1950. The reconstruction is spliced to the model's projection at 2025, where both are indexed to 1. Method and sources are in the script header.



Human capital at constant cost, 1925-2100, by region

World

Year	Population 25-64 (M)	Entrants (M/yr)	Tertiary share of 25-64	Entrants / population	Investment / GDP	Gross stock / GDP	Constant-cost stock (2025 = 1)
1925	816	37	1%	1.9%	9.3%	2.2	0.18
1950	1,042	44	1%	1.8%	9.5%	2.4	0.24
1975	1,575	72	3%	1.8%	11.1%	2.6	0.39
2000	2,778	104	9%	1.7%	10.7%	3.3	0.71
2010	3,342	124	12%	1.8%	11.5%	3.5	0.86
2025	4,083	125	13%	1.5%	10.3%	3.7	1.00

The world's stock of pre-workforce human capital grew between five- and sixfold over the century, 1.7% a year: 1.6 points of that is more people of working age and 0.1 points is more education per person. Growth accelerated through each quarter-century, from 1.3% a year in 1925-1950 to a peak of 2.5% in 1975-2000, then slowed to 1.4% in 2000-2025 and 0.8% in the last decade. On a net basis, with each age group written down straight-line over its band's expected working life, the stock grew faster still (2.1% a year since 1950 against 1.9% gross), because the book value of the workforce rose from 30% to 35% of replacement cost as it filled with recent, better-educated entrants.

Regions

Constant-cost stock index (2025 = 1), gross, with the model's projection alongside so the whole 175 years can be read in one row (the 2050-2100 columns preview the forecast section):

Region	1925	1950	1975	2000	2025	2050	2075	2100	1925-2025 (%/yr)	2025-2100 (%/yr)
United States	0.23	0.34	0.51	0.81	1.00	1.30	1.46	1.50	+1.5	+0.5
OECD ex-US	0.31	0.42	0.60	0.90	1.00	1.22	1.38	1.44	+1.2	+0.5
China	0.12	0.15	0.27	0.68	1.00	0.81	0.59	0.43	+2.1	-1.1
India + South Asia	0.10	0.12	0.21	0.51	1.00	1.31	1.44	1.23	+2.3	+0.3
Latin America	0.07	0.12	0.25	0.57	1.00	1.05	1.02	0.86	+2.7	-0.2
SE Asia + Pacific	0.07	0.11	0.22	0.53	1.00	1.10	1.13	0.97	+2.7	-0.0
Russia + CIS	0.24	0.33	0.59	0.90	1.00	0.94	0.84	0.72	+1.4	-0.4
MENA	0.06	0.08	0.15	0.42	1.00	1.58	1.99	1.94	+2.9	+0.9
Sub-Saharan Africa	0.07	0.10	0.19	0.43	1.00	2.22	3.45	3.84	+2.7	+1.8
World	0.18	0.24	0.39	0.71	1.00	1.16	1.23	1.18	+1.7	+0.2

Every region but one built human capital in every quarter-century of the last hundred years; the exception is Russia + CIS, whose series falls in 1940-1945 and again after 2020. China's reversal is the sharpest in the table: from 3.7% a year in 1975-2000 (second only to MENA's 4.2%), and still 1.5% a year in 2000-2025, to a 1.1% a year decline. Russia + CIS has already stalled, with its 2025 stock below 2015, and the OECD ex-US has been flat for a decade. The regions that grew fastest in the past century (MENA, Latin America, SE Asia, Sub-Saharan Africa, all between 2.7% and 2.9% a year) are the ones with the most building left, and only Sub-Saharan Africa keeps anything like its historical pace.

The same splice on a net basis (Barro-Lee's 10-year age groups from 1950, cohort-shifted after 2015) tracks the model's own net stock:

Region	1950	1975	2000	2025	2050	2075	2100	Net / gross 1950	Net / gross 2025
United States	0.31	0.51	0.84	1.00	1.33	1.41	1.43	0.35	0.38
OECD ex-US	0.34	0.59	0.98	1.00	1.23	1.35	1.39	0.27	0.33
China	0.12	0.26	0.75	1.00	0.72	0.54	0.39	0.28	0.34
India + South Asia	0.10	0.17	0.49	1.00	1.53	1.43	1.20	0.30	0.37
Latin America	0.10	0.22	0.60	1.00	1.10	1.01	0.84	0.31	0.35
SE Asia + Pacific	0.10	0.21	0.54	1.00	1.19	1.14	0.94	0.31	0.35
Russia + CIS	0.26	0.53	0.91	1.00	0.92	0.82	0.68	0.30	0.38
MENA	0.06	0.12	0.42	1.00	1.89	2.00	1.89	0.29	0.38
Sub-Saharan Africa	0.08	0.15	0.40	1.00	2.82	3.64	3.89	0.30	0.38
World	0.20	0.37	0.75	1.00	1.19	1.20	1.14	0.30	0.35

The OECD ex-US has been flat on a net basis since 2000 and Russia + CIS nearly so. The model's net stock in 2025 is 44% of gross, above the reconstruction's 35%, because it prices the in-service workforce rather than everyone aged 25-64. The two series agree on direction and turning points, so the splice does not depend on the gross/net choice.

The United States

Year	Entrants (M/yr)	Entrants / population	Entrant tertiary share	Workforce 25- 64 (M)	Tertiary share of 25- 64	Investment / GDP	Constant-cost stock (2025 = 1)
1925	1.90	1.70%	8%	53	5%	10.7%	0.23
1940	2.17	1.67%	10%	65	6%	11.2%	0.28
1950	2.35	1.53%	13%	77	8%	10.4%	0.34
1960	2.22	1.23%	16%	84	10%	8.8%	0.39
1970	3.47	1.67%	25%	92	14%	12.7%	0.45
1980	4.35	1.89%	32%	110	22%	14.8%	0.58
1990	3.86	1.52%	33%	131	28%	12.1%	0.71
2000	3.89	1.38%	33%	148	29%	10.9%	0.81
2010	4.46	1.44%	41%	166	33%	11.7%	0.93
2020	4.45	1.31%	40%	178	33%	10.7%	1.00
2025	4.61	1.34%	40%	178	33%	11.0%	1.00

The US series has one large hump: the baby boom entering the workforce with the first mass college cohorts, which took investment from under 9% of GDP in 1960 to nearly 15% in 1980. Since then the share has drifted around 11 to 12%, more education per entrant offsetting a falling youth share. The stock grew 1.5% a year over the century (1.2 points people, 0.3 education), but only 0.3% a year over 2015-2025, and the 25-64 population has been flat since 2020. The reconstruction reaches 2025 with the US stock already at a plateau.

Spending

Investment in new entrants as a share of GDP:

Region	1925	1950	1975	2000	2010	2025
United States	10.7%	10.4%	13.6%	10.9%	11.7%	11.0%
OECD ex-US	9.8%	9.4%	10.4%	9.8%	9.2%	8.3%
China	6.8%	8.6%	12.1%	10.5%	13.6%	9.4%
India + South Asia	6.1%	7.5%	8.7%	11.1%	12.0%	12.5%
Latin America	8.3%	8.6%	10.1%	11.8%	12.0%	11.5%
SE Asia + Pacific	7.7%	8.3%	9.3%	11.6%	11.9%	11.6%
Russia + CIS	14.6%	13.8%	12.8%	12.5%	14.7%	9.4%
MENA	7.2%	7.2%	8.7%	11.9%	13.0%	12.0%
Sub-Saharan Africa	7.4%	7.6%	8.2%	9.8%	10.3%	12.1%
World	9.3%	9.5%	11.1%	10.7%	11.5%	10.3%

The world spends slightly more of its output on new human capital than it did a century ago, and less than it did fifteen years ago. The share rose from about 9% of GDP in 1925 to a peak of 11.7% in 1980-85 as the baby-boom cohorts entered with the first mass tertiary intakes, stayed near 11% through 2010 as the education mix upgraded faster than cohorts shrank, and has since fallen back to about 10% as the youth share of population dropped from 1.8% to 1.5%. Per entrant, the world spends 1.6 times its 1925 multiple of GDP per capita on rearing and schooling (4.4 to 7.2 times GDP per capita), and GDP per capita is itself far higher.

Outside data say the same. Total US education spending has held at about 5.4 to 5.6% of GDP in recent years, having peaked at 6.7% in 2009 (World Bank/UIS series), while the US birth cohort fell from 4.32M in 2007 to 3.63M in 2024 (NCHS). World government education spending was 4.16% of GDP in 2019 and 3.51% in 2023 (World Bank, from UNESCO UIS). The ledger's 10% is not comparable to those 4 to 6% education shares: it also counts rearing to entry age. With rearing excluded the model's 2025 flow is 4.1% of GDP, close to the world education share.

Joining the series

The two series join at 2025 with one visible seam: the reconstruction's entrants are the cohort now aged 20-24 (125M worldwide, 4.6M in the US), while the model's entrant flow is the cohort aged 0-19 divided by 20 (119M and 4.1M). The model's figure is the average of the next twenty years of entrants and is the right one for the projection; the reconstruction's is the right one for the present. Where births have been falling, as in the US since 2007, the two differ by the amount of that decline, which is why the reconstruction's 2025 investment share (10.3% of world GDP, 11.0% for the US) sits above the model's (9.6% and 11.4%). The tertiary shares differ for a reason met already in China: the reconstruction's 13% is completed tertiary attainment among everyone aged 25-64 from Lee-Lee, while the model's 24% is the tertiary-plus share of the in-service workforce from national college-attainment statistics.

Forecast, 2025-2100

Two nets

The projection needs one more definition. Each region books immigrants at its own replacement cost on arrival (the migration transfer) and then depreciates them alongside everyone else; the origin writes them down at its own, lower cost. So a region's investment less its total charge is not an own-cohort measure: a steady receiver drifts into deficit simply because its immigrant stock accumulates and wears down. The ledger tracks post-2025 arrivals as a subset of each region's workforce and reports two nets: **own-cohort net**, investment less the charge on the region's own cohorts, and **total net**, which adds the migration transfer and subtracts the charge on immigrants. At the world level the two coincide (every migrant is somebody's own cohort), and the gap between the regional sums is the transfer itself: the same people re-priced at destination cost.

World

Year	Investment \$T	Depreciation + write-offs \$T	Net \$T	Investment / GDP	Constant-cost net stock (2025 = 1)	Entrants (M)	Tertiary+ share of workforce
2025	15.2	15.2	-0.0	9.6%	1.00	119.2	24%
2030	17.1	16.8	+0.3	9.7%	1.04	119.6	27%
2040	22.6	22.6	0.0	9.6%	1.13	118.8	37%
2050	30.6	32.2	-1.6	9.2%	1.19	116.3	45%
2060	40.7	44.9	-4.2	8.9%	1.22	112.8	52%
2075	60.7	70.6	-9.8	8.4%	1.20	106.7	59%
2100	144.4	170.4	-26.0	7.8%	1.14	95.7	59%

The constant-cost stock rises while net investment is zero or negative, and it is worth being exact about why. Its change from 2025 splits into three lines, each deflated by its region's cost index, and they sum to the change without residual:

Cumulative since 2025, \$T at 2025 cost	2030	2040	2050	2060	2075	2100
Net investment (investment less all charges)	+1.3	+4.7	+1.3	-11.0	-43.0	-104.4
Migration transfers (movers re-booked at destination cost)	+6.7	+20.7	+35.5	+50.8	+73.9	+112.0
Useful-life revaluation (longer working lives raise remaining book value)	+1.4	+4.3	+7.3	+10.6	+15.8	+24.2
Change in the constant-cost stock	+9.4	+29.8	+44.2	+50.3	+46.7	+31.8

Three things stand out.

1. **The world is at break-even now, and the build is ending.** At current cost the ledger capitalizes exactly what it charges in 2025 and hovers within half a trillion dollars of zero until 2040, then runs a growing net disinvestment. The constant-cost stock nevertheless rises about 0.8% a year through the early 2030s and half that by the 2040s, peaks in 2063 at 22% above 2025, and drifts down about 0.2% a year to end 14% above. Almost none of that is new human capital: the table above puts it in migration transfers and lengthening working lives. The migration line is the same people re-priced: a graduate who moves from India to the OECD is written down at Indian cost and written up at OECD cost, and the world stock rises by the difference. With migration switched off the world constant-cost stock peaks at 1.09 in the 2050s and ends the century at 0.92. The honest world reading is flat to slightly down on people and education, with the reported rise coming from where the people are priced.
2. **What is being built is education, not headcount.** Entrants fall from 119M to 96M a year, but the tertiary-plus share of the in-service workforce goes from a quarter to three-fifths, and a tertiary entrant carries about twice the cost of a secondary one in the world average (\$204k against \$98k in 2025), because tertiary entrants are concentrated in richer regions. Nearly all of the quantity growth to 2060 is composition; from the 2080s the tertiary headcount is itself falling.
3. **The spending share slips, not collapses.** Investment runs 9.6% of GDP now and 7.8% in 2100, because entrant cohorts shrink relative to GDP while per-entrant cost tracks income. Spending per entrant rises throughout; spending on entrants as a whole loses under two points of GDP over the century, continuing the post-2010 slide in the reconstruction.

Regions

Constant-cost net stock index (2025 = 1), the peak year of that index, the first years own-cohort and total net investment are negative, and the change in annual entrants over the century. The world is at break-even in 2025 (-\$0.01T) and first runs a sustained deficit in 2041; at the world level the two nets coincide.

Region	2030	2050	2075	2100	Peak	Peak level	Own cohorts < 0	Total < 0	Entrants 2100 / 2025
United States	1.09	1.33	1.41	1.43	2100	1.43	2064	2065	0.95
OECD ex-US	1.04	1.23	1.35	1.39	2100	1.39	2025	2086	1.11
China	0.93	0.72	0.54	0.39	2025	1.00	2025	2025	0.36
India + South Asia	1.14	1.53	1.43	1.20	2056	1.56	2061	2056	0.56
Latin America	1.02	1.10	1.01	0.84	2053	1.11	2068	2051	0.56
SE Asia + Pacific	1.03	1.19	1.14	0.94	2058	1.21	2072	2058	0.61
Russia + CIS	0.97	0.92	0.82	0.68	2025	1.00	2025	2025	0.59
MENA	1.22	1.89	2.00	1.89	2065	2.01	2062	2063	0.84
Sub-Saharan Africa	1.37	2.82	3.64	3.89	2100	3.89	never	never	1.26
World	1.04	1.19	1.20	1.14	2063	1.22	2041	2041	0.80

The regions fall into four groups.

- **Drawing down from today: China, Russia + CIS.** China charges more depreciation than it capitalizes in every year of the run. Its entrant cohort falls by nearly two-thirds and its constant-cost stock by 61% by 2100, even though the rising college share means each entrant embodies more. In current dollars the Chinese stock still rises more than 10-fold, which is the revaluation trap: a shrinking, better-educated workforce repriced at higher income looks like accumulation. Russia + CIS follows the same shape at a gentler slope. The century ahead is the first in which any region draws down by choice of fertility rather than by catastrophe.
- **Building to the 2050s, then drawing down: India + South Asia, Latin America, SE Asia + Pacific, MENA.** These are the demographic-dividend regions. India peaks in 2056 at 1.56 times its 2025 stock and gives back most of the gain by 2100. MENA doubles and holds. India, Latin America, and SE Asia are net exporters of trained people, written down at origin cost as they leave, and the outflow brings their turning points forward by 5 to 17 years; MENA is a small net receiver and its two turning points nearly coincide.
- **Building throughout: Sub-Saharan Africa.** The only region whose entrant cohort is larger in 2100 than in 2025, and the only one whose own-cohort net investment never turns negative. Its stock nearly quadruples, from a base that is 3% of the world total.
- **Building on their own cohorts and on immigration, then on neither: United States, OECD ex-US.** Both rich-region stocks rise through 2100. The US covers its own charge with a small surplus until 2064; the OECD ex-US starts in own-cohort deficit (its opening workforce carries larger vintages than its 7M-a-year entrant flow replaces) and moves into surplus by the 2040s as those vintages retire. On top of that, working-age arrivals booked at the destination's replacement cost add \$0.6T a year to the US and \$1.1T to the OECD ex-US in 2025, rising with income. Those arrivals are then depreciated where they landed, and by 2075 (US)

the charge on them has grown to the size of the transfer that brought them in; the US section below has the flows. Total net investment turns negative in 2065 (US) and 2086 (OECD ex-US).

The United States

Year	Entrants (M)	Investment \$T	Charge \$T	of which on post- 2025 immigrants	Own- cohort net \$T	Migration transfer \$T	Total net \$T	Investment / GDP
2025	4.1	2.72	2.54	0.03	+0.20	0.63	+0.81	11.4%
2030	4.0	2.88	2.84	0.16	+0.20	0.69	+0.73	11.0%
2040	4.0	3.55	3.85	0.49	+0.19	0.90	+0.60	10.4%
2050	4.0	4.63	5.44	0.96	+0.15	1.22	+0.41	10.0%
2060	4.0	5.88	7.29	1.49	+0.08	1.57	+0.15	9.7%
2075	3.9	8.64	10.99	2.31	-0.04	2.30	-0.04	9.3%
2100	3.9	21.60	27.58	5.78	-0.20	5.59	-0.38	8.8%

The US invests a larger share of GDP than any region in 2025 except MENA and Sub-Saharan Africa, because its entrants are expensive (a college share above 40% and the world's highest GDP per capita) rather than numerous. Its entrant cohort is flat at about 4.0M a year for the whole century, held up by a 1.4 fertility floor and by immigration.

The US covers the charge on its own cohorts with a surplus of about \$0.2T a year through mid-century. What it does not do is grow on its own cohorts: the surplus is 8% of the charge in 2025 and shrinks to zero by 2064, as the flat entrant flow is set against a stock whose depreciation rises with income. The migration transfer is worth about a fifth of gross additions throughout, and it is what makes the total net clearly positive to the 2050s. It also builds a stock that has to be written down: the charge on post-2025 immigrants goes from \$0.03T in 2025 to \$2.3T in 2075, where it equals the transfer; the total net crosses zero in 2065. With migration switched off everywhere, which also removes migrants' children from future cohorts (-- `set=demographics.migrationMultiplier=0`), US entrants fall by 35% instead of 5%, own-cohort net turns negative in 2043, the US constant-cost stock peaks in 2046 at 1.07 and ends the century at 0.88, and the OECD ex-US stock declines from 2025 to 0.76 by 2100. The reconstruction reached 2025 with the US stock at a plateau; the model's 1.1% a year growth to 2050 rests on its immigration assumption and on entrants' college share continuing to rise. The US result is therefore an immigration-policy result at least as much as a demographic one: on its own children it holds level; the growth is immigrants and their children.

The size of that immigration assumption deserves stating. The demographics module's US migration rate is calibrated to net immigration of about 1.2M a year (UN WPP 2024 and CBO 2025), but the model closes the world's migration by scaling every receiver's inflow to the emigration budget, so the flow the ledger actually books is 1.6M a year, 1.27M of them working-age; the OECD ex-US receives 4.3M rather than the 3M its rate was set to. Both are about a third above the figures they were calibrated to, so the rich-region transfers above are an upper bound.

Robustness

- **Energy and climate scenarios do not move the quantity path.** ssp3-70 and net-zero produce identical constant-cost stocks, entrant flows, and regional peak years to the default run; only the dollar values differ, because those scenario files leave demographics untouched. The human-capital trajectory in this model is a demographic and education result, not an energy one.
- **Cost scope shifts levels and the world turning point, not the regional ordering.** Adding students' foregone earnings at the Kendrick/BEA share (foregoneEarningsShare 0.45) raises the 2025 flow to 11.9% of GDP and keeps the world's net investment positive until 2046, because the opportunity cost weights the young, tertiary-heavy cohorts; pricing rearing at the National Transfer Accounts midpoint (0.30) gives 11.3% and leaves the turn at 2025; removing rearing gives 4.1% and a turn in 2047. Peak years move by at most three years in every case (Latin America, 2053 to 2056, with rearing removed), the US own-cohort turn stays at 2064, and the four regional groups are unchanged.

Conclusion

The world spent a century building human capital at an accelerating pace and has now stopped. The reconstruction shows the stock of people times education growing in every quarter-century since 1925 in every region but one, fastest in 1975-2000, and slowing since; the projection shows the ledger at break-even in 2025, net investment negative from the 2040s, and a constant-cost stock whose reported rise to the 2060s is the re-pricing of migrants and the lengthening of working lives rather than new human capital. The result is demographic and educational, not energy or climate related: the energy scenarios leave it untouched, and the cost-scope sensitivities move levels, not turning points or ordering.

Three readings follow.

- **Composition is doing the work.** Entrants fall while the tertiary share of the workforce more than doubles, so the stock keeps its value on fewer, costlier people until the 2060s and then loses it. Any policy question about "investing in human capital" is, on this measure, a question about education per entrant, because the number of entrants is set.
- **The rich regions' growth is immigration.** The US holds level on its own cohorts into the 2060s and grows because it books working-age arrivals at its own replacement cost; the OECD ex-US grows the same way on a larger flow, and the model's flows are an upper bound. The source regions write the same people down at their own cost, and the world's reported gain is the price difference.
- **Spending is not the constraint.** The world has put 9 to 12% of GDP into new entrants for a hundred years; the projected slide is entrant cohorts shrinking relative to output, with spending per entrant rising throughout.

What the result does not say:

- Cost is not value. A lifetime-income (Jorgenson-Fraumeni) account would be several times larger and would respond to wage premia this ledger ignores.
- Post-entry training, learning on the job, and obsolescence are outside the ledger; a shrinking pre-workforce stock is compatible with rising skill if on-the-job investment grows.
- The 2025 workforce is seeded, not observed. The ledger starts each region from demographics' working-age headcount spread uniformly over the 45 working ages and thinned by the survival curve, because the model carries no age structure inside the working cohort. That seed sets every region's 2025 charge and the timing

of its own-cohort turn: the OECD ex-US and China start in deficit because their seeded vintages (10.3M and 21M a year) are larger than their entrant flows (7.2M and 12M), and the US starts in surplus because its seed (4.4M) is close to its flow (4.1M). Real age structures are lumpier (the 1960s cohorts now retiring in the OECD, China's 1960s-80s cohorts), so the direction is right and the timing is approximate.

- Immigrants are booked at the destination's replacement cost written down for tenure (about three-quarters of full cost), with no under-employment discount and a 70% college share, so the rich-region migration transfers are an upper bound; the appendix sizes the difference. The immigrant subset the ledger tracks starts empty in 2025: people who arrived before the run are own cohorts to it, so "own-cohort" means post-2025 births, not nativity.
- Longer working lives raise the book value of everyone already at work. The ledger books that revaluation on its own line so the accounts close; it is a schedule change, not investment.
- The reconstruction prices everyone aged 25-64, not the in-service workforce: the model nets out domestic-role and disability exits, the reconstruction cannot, because participation by age and education is not observed before the 1990s. That is a level difference (the net/gross ratios above), and where participation rose over the century, as women's did in the OECD after 1950, the in-service stock grew faster than the series shows. Migration is inside the population counts, so the stock is complete, but the flows are not split between births and arrivals.
- China's tertiary shares are census and enrollment based from 2000 on, a correction to the source rather than an observation of attainment by age; the Barro-Lee age profile is scaled to the census total. Before 1950 the covered countries are scaled to regional population totals, which assumes uncovered countries had the covered average attainment; Russia + CIS and India before 1950 rest on Lee-Lee's Soviet-era and British-India series, used only as ratios, and should be read as rough.
- Demographics follow the UN WPP 2024 low variant, so entrant cohorts outside Sub-Saharan Africa shrink faster than a medium-variant path would give; the medium variant would delay every peak year but not remove the ordering.

Appendix: cross-check against the G7-BRIC spreadsheet

The G7-BRIC human-capital model (v4, 4 September 2026) is a single-year 2025 account for eleven countries in billions of 2024 PPP dollars, with five or six education bands per country. It capitalizes paid care and schooling only (no student opportunity cost, no unpaid care), applies participation at entry, amortizes straight-line over effective exit age minus entry age (41-45 years in the G7, 32-42 in the BRICs) with WHO age-specific mortality and no other pre-retirement exits, and values migrants at remaining book value at age 32 with the domestic education mix. The ledger's default cost scope matches it (its US care cost through 18 is 23% of GDP per capita, the USDA figure the ledger uses). The two models agree on every sign and on the ordering of countries; what remains between them is the participation filter, the treatment of working life, and migration.

2025 flows in \$T at current cost, and the end-2025 net stock. Model regions are wider than the spreadsheet's countries: India + South Asia, Russia + CIS, and OECD ex-US (all Europe, Japan, Korea, Canada, Australia, New Zealand, Israel) against India, Russia, and the six non-US G7 members.

	Entrants (M)	Investment	Depreciation + write-offs	Own- cohort net	Migration	Net incl. migration	Net stock
United States: spreadsheet	3.64	2.41	2.42	-0.01	+0.36	+0.35	55.7
model, default	4.08	2.72	2.54	+0.20	+0.63	+0.81	42.9
model, life = exit minus entry	4.08	2.72	2.54	+0.20	+0.65	+0.83	55.0
China: spreadsheet	11.97	1.80	1.90	-0.10	-0.01	-0.11	38.0
model, default	11.99	2.34	3.32	-0.98	-0.04	-1.01	54.2
model, life = exit minus entry	11.99	2.34	3.34	-1.00	-0.04	-1.04	72.1
India: spreadsheet	17.0	0.87	0.70	+0.17	-0.01	+0.16	13.6
model (India + South Asia), default	31.5	1.78	1.35	+0.43	-0.09	+0.35	14.8
model, life = exit minus entry	31.5	1.78	1.34	+0.44	-0.09	+0.35	29.5
Russia: spreadsheet	1.31	0.30	0.42	-0.12	-0.02	-0.14	7.5
model (Russia + CIS), default	3.19	0.68	0.74	-0.06	-0.04	-0.10	12.1
model, life = exit minus entry	3.19	0.68	0.74	-0.06	-0.04	-0.10	16.1
G7 ex-US: spreadsheet (six countries)	3.96	1.69	2.08	-0.39	+0.19	-0.20	44.2
model (OECD ex-US), default	7.20	3.08	3.80	-0.67	+1.10	+0.39	61.5
model, life = exit minus entry	7.20	3.08	3.80	-0.69	+1.15	+0.43	82.1

"Life = exit minus entry" is the model rerun with `npm run human-capital:trajectory -- --no-exit-hazards`, which switches off death, disability, and domestic-role exits so that useful life is retirement age minus entry age and there are no write-offs, as in the spreadsheet. In 2025 the model's own-cohort net excludes only the one-year charge on that year's arrivals (\$0.03T for the US), so it is comparable to the spreadsheet's net before migration.

The model's \$2.72T of 2025 US investment becomes the spreadsheet's \$2.41T in one step: the spreadsheet applies participation at entry (3.64M effective entrants against the model's 4.08M), which the model handles instead through exit hazards and write-offs over the career. Depreciation matches at \$2.5T against \$2.4T. The net stock matches at \$55T once working life is defined the same way; on the model's own shorter expected working lives the US stock is \$43T. Both accounts put the US at about break-even on its own children in 2025 (+\$0.20T against -\$0.01T) and clearly positive only because of immigration. China, Russia and the non-US G7 are net disinvestors on their own cohorts in both; India is a net investor in both. The China entrant count is identical (12.0M), because both take the cohort turning 20 from the same UN age structure, and the spreadsheet's entrant

tertiary share for China (55%) is close to the model's 61%, both far above the Lee-Lee figure the reconstruction had to override.

Where they differ:

1. *Care cost outside the US*. The spreadsheet scales care by household consumption relative to the US rather than by GDP per capita, which roughly halves it for China (China's consumption share of GDP is about 38% against the US's 68%); this is most of the gap in China's investment (\$1.8T against \$2.3T) and stock (\$38T against \$54T).
2. *Useful life and exits*. The spreadsheet amortizes over 41-45 years and charges only mortality. The model's expected working lives are 28-39 years because they net out disability and domestic-role exits, which it writes off at book value. That is why the model's charge exceeds the spreadsheet's by more than its investment does, and why its own-cohort net is more negative for China (-\$1.0T against -\$0.1T). The spreadsheet's note that net "still excludes unestimated disability and unscheduled retirement" and its break-even sensitivity (a 0.16% annual non-fatal loss would erase the G7's net) point at the same mechanism from the other side. A third candidate for the China gap is the model's opening age structure (see the conclusion's limits), which the spreadsheet takes from observed cohorts.
3. *Migration*. This is the one substantive disagreement. The spreadsheet's G7 ex-US is a net disinvestor even after migration (-\$0.20T); the model's OECD ex-US turns positive (+\$0.39T). Two assumptions drive it. The model's demographics move 3.4M net working-age migrants a year into the OECD ex-US and 1.27M into the US, against the spreadsheet's 1.1M and 0.73M effective (1.3M times a 70% working-age share and 80% participation). And the model books migrants with a 70% college share at the destination's replacement cost written down for about ten years of tenure, about \$500k each for the US, against the spreadsheet's domestic mix at remaining value at age 32, about \$500k on its lower unit costs but for far fewer people. Together they make the model's migration transfer 1.8 times the spreadsheet's for the US and 6 times for the OECD ex-US. The spreadsheet's treatment is the more conservative and the better documented; the model's migration flow and college-share assumptions are the two places in this note where the reader should discount.
4. *Scope of the regions*. The model's India + South Asia has 1.9 times the spreadsheet's Indian entrants, and Russia + CIS 2.4 times Russia's, because Pakistan, Bangladesh and Central Asia are young; the regional Russia net is less negative than the spreadsheet's Russia. The spreadsheet's country figures are the better read of Russia itself.

None of the note's conclusions turn on the level. The peak years, the regional ordering, and the US dependence on immigration come from demographics and the education mix, which the two models share. What the spreadsheet changes is the size of the immigration offset for the rich regions: on its conventions, the US stays marginally positive and the rest of the G7 does not.

Sources

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- *G7-BRIC human-capital model v4* (Google Sheets, 4 September 2026): 2025 flows for the G7 and BRICs from World Bank cohorts, OECD Education and Pensions at a Glance, WHO life tables, and national migration statistics.
- docs/HUMAN_CAPITAL.md for the ledger's method, calibration, and prior art (Kendrick 1976; Eisner 1985; Mallatt 2026; Eurostat duration of working life).